Drone design, fabrication and programming

Parts, integration, commissioning, batteries, configurations, software and verification (Appendix 4B)

Classification and parts

- Drone Rules 2021: nano ≤ 250 g, micro > 250 g–2 kg, small 2–25 kg, medium 25–150 kg, large > 150 kg.
- Four groups: structure; propulsion (motor, ESC, propeller); avionics (FC, sensors, radio); power (battery, PDB, BEC).
- KV = rpm per volt, no load. Prop = diameter × pitch. Adjacent rotors counter-rotate.
- ESC: PWM needs endpoint calibration; DShot is digital with checksum, no calibration.

Integration and electronics

- UART point-to-point; I²C 2-wire shared bus with addresses; SPI fast with chip selects; CAN differential, multi-node, robust.
- $T_s = 1/f_s$ (400 Hz → 2.5 ms). ADC: 2^n codes, LSB = $V_{ref}/2^n$, ideal SNR = 6.02N + 1.76 dB. Data rate = N × f_s.
- One time base for all logs. Soft-mount the FC, twist power pairs, keep the compass away from high current.

Commissioning order

- Incoming inspection → assembly → CG → wiring (polarity, continuity, shorts, smoke stopper) → firmware and parameters → calibration → motor order and direction, props off → pre-arm → tethered hover → acceptance test.

Battery equations

- Series adds V; parallel adds Ah.
- E [Wh] = V_{nom} × Ah; I_{max} = C × Ah; sag V = V_{OC} − IR.
- E_{usable} = E × DoD × k_{temp} × k_{age} − reserve; endurance t = E_{avail} / P.
- LiPo cell: 4.2 V full, 3.7 V nominal, 3.3–3.5 V minimum under load; store at about 3.8 V.

Configurations and control

- T/W ≥ 2 for normal multirotors.
- Quad-X mixing: motor = throttle ± roll ± pitch ± yaw; saturation removes authority.
- Thrust square law $F = k_1\omega^2$.

Software and verification

- Layers: firmware → parameters → mission planning → high-level APIs, joined by MAVLink.
- State machine: explicit states, guards, prioritized failsafes; reject invalid transitions.
- Discrete PID: filtered derivative, output limits, anti-windup.
- Unit → integration → SIL → HIL → bench → tethered → progressive flight. One configuration ID ties hardware, firmware, parameters and calibration.

Exam traps

- Series raises Ah, or parallel raises V (both wrong).
- Calling a waypoint mission "autonomous"; it is automatic within limits.
- Props on before the motor-direction test; raising gains to hide vibration; forgetting the reserve.

Source: course reader, Appendix 4B